

# Installing and Using Node-RED on Ubuntu



Node-RED adds to the benefits of Node.js. This article shows how to install and use it on the popular open source operating system Ubuntu.

**N**ode-RED is a hardware device wiring tool that uses programming. Different nodes of devices, hardware and other APIs can be wired together in a browser based tool using Node-RED. It has become very important software in the field of IoT as it connects APIs, hardware devices and online services in new and interesting ways. This lightweight tool has been built on Node.js because of the amazing non-blocking and event-driven model it provides. It can be said that Node-RED basically acts as a controller for IoT devices. Hence, it is advisable for people interested in IoT to learn more about this software.

The first step is to install Node.js, since Node-RED is developed on it. We do that by using the following command, as shown in Figure 1.

```
sudo apt-get install nodejs-legacy
```

Next, to check if our installation has taken place properly, we use the following command, as shown in Figure 2.

```
Node -v
```

```
jishnusaaurav@jishnusaaurav-IdeaPad-L340-15IRH-Gaming:~$ sudo apt-get install nodejs-legacy
[sudo] password for jishnusaaurav:
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'nodejs' instead of 'nodejs-legacy'
The following packages were automatically installed and are no longer required:
  libevent-core-2.1-6 linux-hwe-5.4-headers-5.4.0-73
  linux-hwe-5.4-headers-5.4.0-74 linux-hwe-5.4-headers-5.4.0-77
  linux-hwe-5.4-headers-5.4.0-80 linux-hwe-5.4-headers-5.4.0-81
  linux-hwe-5.4-headers-5.4.0-84 linux-hwe-5.4-headers-5.4.0-86
  linux-hwe-5.4-headers-5.4.0-87
Use 'sudo apt autoremove' to remove them.
The following packages will be upgraded:
  nodejs
1 upgraded, 0 newly installed, 0 to remove and 81 not upgraded.
Need to get 25.1 MB of archives.
After this operation, 1,289 kB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_14.x bionic/main amd64 nodejs amd64 14.18.1-1nodesource1 [25.1 MB]
Fetched 25.1 MB in 15s (1,672 kB/s)
N: Ignoring file 'nvi' in directory '/etc/apt/sources.list.d/' as it has no filename extension
(Reading database ... 579459 files and directories currently installed.)
Preparing to unpack .../nodejs_14.18.1-1nodesource1_amd64.deb ...
Detected old npm client, removing...
Unpacking nodejs (14.18.1-1nodesource1) over (14.17.1-1nodesource1) ...
Setting up nodejs (14.18.1-1nodesource1) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
N: Ignoring file 'nvi' in directory '/etc/apt/sources.list.d/' as it has no filename extension
```

Figure 1: Node.js installation

```
jishnusaaurav@jishnusaaurav-IdeaPad-L340-15IRH-Gaming:~$ npm -v
6.14.5
jishnusaaurav@jishnusaaurav-IdeaPad-L340-15IRH-Gaming:~$
```

Figure 2: Checking the node version

```
jishnusaaurav@jishnusaaurav-IdeaPad-L340-15IRH-Gaming:~$ sudo npm install -g --unsafe-perm node-red
node-red-admin

added 351 packages, and audited 351 packages in 28s

28 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 7.0.14 -> 8.1.4
npm notice Changelog: https://github.com/npm/cli/releases/tag/v8.1.4
npm notice Run npm install -g npm@8.1.4 to update!
npm notice
jishnusaaurav@jishnusaaurav-IdeaPad-L340-15IRH-Gaming:~$
```

Figure 3: node-red-admin installation